

System Requirements

This document outlines the specific system requirements for installing and running the ALTA MODBUS TCP Utility. Meeting these requirements is essential for a successful installation and for ensuring the application runs reliably and securely.

Operating System

Windows The application is designed for modern, 64-bit Windows environments.

- Supported: Windows 10 (64-bit, v1607+), Windows 11 (64-bit), Windows Server 2016/2019/2022.
- Unsupported: 32-bit Windows, Windows 8.1, Windows 7.

macOS

- Supported: macOS 10.15 (Catalina) or newer.
- Architecture: Intel (x64) or Apple Silicon (M1/M2/M3 via Rosetta 2 or native arm64 build).

Linux

- Supported: Modern distributions (Ubuntu 20.04+, Debian 10+, Fedora 32+).
- Dependencies: Standard desktop libraries (libnotify, libappindicator, etc.) required for Electron apps.

Software Dependencies

The installer is designed to handle the primary software dependency automatically.

Node.js (JavaScript Runtime):

Required Version: v22.11.0 (LTS) or a compatible v22.x release.

Installation: The installer will automatically check for Node.js. If it is not found, the installer will download and install the required version silently in the background. No separate installation is required by the user.

Web Browser:

A modern, up-to-date web browser is required to access the user interface.

Recommended Browsers: Google Chrome, Microsoft Edge, Mozilla Firefox (latest versions).

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For the best experience, including the ability to "install" the application to the desktop and Start Menu, a Chromium-based browser (Chrome or Edge) is recommended.

Unsupported: Internet Explorer is not supported.

Hardware Requirements

Processor (CPU):

Minimum: 1 GHz 64-bit (x64) processor

Recommended: 2 GHz dual-core 64-bit (x64) processor or better

Memory (RAM):

Minimum: 2 GB

Recommended: 4 GB or more

Disk Space:

Application: Approximately 300 MB for the application and all dependencies.

Data Storage: A minimum of 1 GB of free disk space is recommended for data logging.

User Permissions

Installation: The installer must be run with Administrator privileges (Windows) or sudo (Mac/Linux) if installing to system directories.

Runtime: The user does not need Administrator privileges for day-to-day use, except potentially to bind to privileged ports (like SNMP Port 162 on Linux/Mac).

Network & Port Requirements

To ensure full functionality, the following network ports must be open and available.

Local Ports (Inbound): These ports are used by the application to listen for data. Ensure Windows Firewall allows inbound traffic on these ports if you use the respective features.

- **TCP 7001:** (Required) Local web server. Used to display the dashboard.
- **UDP 162:** (Optional) SNMP Trap Listener. Required only if using the SNMP Trap plugin. *Note: On Linux/Mac, ports < 1024 require root privileges.*
- **UDP 1700:** (Optional) LoRaWAN Listener. Required only if using the LoRaWAN plugin.

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- **UDP 47808:** (Optional) BACnet/IP. Required for BACnet discovery and server functionality.

Remote Ports (Outbound): These ports are used by the application to connect to external devices and services.

- **TCP 502:** (Required) MODBUS TCP. Used to connect to Monnit Gateways.
- **TCP 80:** (Required) HTTP. Used to fetch Gateway configuration pages.
- **TCP 1883 / 8883:** (Optional) MQTT. Used to connect to an MQTT broker (1883 for unencrypted, 8883 for TLS).
- **TCP 443:** (Optional) HTTPS. Used for Webhook broadcasts (if using HTTPS endpoints) and sending SMS alerts via Textbelt API.
- **TCP 587 / 465 / 25:** (Optional) SMTP. Used for sending Email alerts (port depends on your email provider).

Firewall Configuration:

The firewall must allow outbound TCP traffic on port 502 to the IP addresses of the Monnit gateways.

Network Connectivity:

The computer running the utility must be on the same local area network (LAN) and IP subnet as the Monnit Ethernet Gateway(s).